

Cryptography— How It Works (Plain-English Guide)

Process



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The Big Idea

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CRYPTOGRAPHY IS THE ART OF LOCKING INFORMATION SO ONLY THE RIGHT PEOPLE CAN UNLOCK IT.

THINK OF WRITING A NOTE, PUTTING IT IN A LOCKED BOX, AND HANDING THE KEY ONLY TO YOUR FRIEND.



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Two Main Locks

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Lock Type	How It Works	Real-Life Analogy
Symmetric (One Key)	Same secret key locks & unlocks. Fast, simple.	One padlock, both of you share the same key.
Asymmetric (Two Keys)	Public key locks, private key unlocks. You publish the public key; keep the private key hidden.	Anyone can drop a letter into your locked mailbox, but only you have the key to open it.

MODERN SYSTEMS OFTEN COMBINE BOTH: USE ASYMMETRIC TO SHARE A QUICK SYMMETRIC KEY, THEN SWITCH TO THE FASTER SYMMETRIC LOCK FOR THE BULK DATA.



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Turning Words Into Code

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A CIPHER IS THE RULE SET THAT SCRAMBLES PLAIN TEXT INTO UNREADABLE TEXT (CIPHER-TEXT).

POPULAR CIPHERS TODAY: AES-256 (SYMMETRIC) AND RSA OR ECC (ASYMMETRIC).



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Keeping It Safe

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- KEY GENERATION – CREATE STRONG RANDOM KEYS (LIKE UNPREDICTABLE DICE ROLLS).
- ENCRYPTION – APPLY THE CIPHER + KEY TO YOUR DATA: PLAIN → CIPHER-TEXT.
- TRANSMISSION / STORAGE – SEND OR STORE THE CIPHER-TEXT; EAVESDROPPERS SEE ONLY NOISE.
- DECRYPTION – RECEIVER APPLIES THE CORRECT KEY + CIPHER TO CONVERT CIPHER-TEXT BACK TO PLAIN TEXT.

IF AN ATTACKER DOESN'T HAVE THE KEY, TODAY'S ALGORITHMS MAKE CRACKING PRACTICALLY IMPOSSIBLE WITHIN HUMAN LIFETIMES.



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Everyday Uses

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- **HTTPS WEBSITES – YOUR BROWSER AND THE SERVER ENCRYPT EVERYTHING YOU READ OR TYPE.**
- **WHATSAPP & SIGNAL – MESSAGES ARE END-TO-END ENCRYPTED; ONLY YOU AND YOUR FRIEND HOLD THE KEYS.**
- **ATM CARDS & UPI – PINS AND TRANSACTIONS TRAVEL LOCKED, SO SKIMMERS CAN'T READ THEM.**
- **DOCUMENT SIGNING – YOUR PRIVATE KEY SIGNS A FILE; ANYONE WITH YOUR PUBLIC KEY CAN VERIFY IT'S REALLY YOURS.**



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Takeaway

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CRYPTOGRAPHY IS YOUR DIGITAL LOCKBOX—
MIXING SMART MATHS WITH SECRET KEYS SO
YOUR CHATS, PAYMENTS AND FILES STAY
PRIVATE, AUTHENTIC AND TAMPER-PROOF.



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